VENKATA SAI GANESH MANDA

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Target Role: Machine Learning and Artificial Intelligence Masters Intern, Apple (Summer 2027)

SUMMARY

Master's student in Robotics and Autonomous Systems (AI) at Arizona State University (GPA 4.0) with hands-on experience building and evaluating production-grade LLM systems, including retrieval-augmented generation, large-scale distributed model training, and automated LLM-judged quality gating, with early exposure to privacy-preserving ML and LLM vulnerability analysis. Proficient in Python, PyTorch, and TensorFlow, with strong foundations in linear algebra, statistics, and optimization.

EDUCATION

M.S. Robotics and Autonomous Systems (AI), Arizona State University — **Aug 2025 – May 2027**
SCAI, Ira A. Fulton Schools of Engineering, Tempe, AZ | GPA: 4.0/4.0
Relevant Coursework: Applied Linear Algebra for Engineers, Robotic Systems I, Agentic AI, Statistical Machine Learning

B.Tech. Artificial Intelligence and Data Science, IIITDM Kurnool — **Sep 2020 – Jun 2024**
Exchange student, University of Agder (UiA), Grimstad, Norway | Jan 2024 – Jun 2024

TECHNICAL SKILLS AND CERTIFICATIONS

Languages: Python, C++, C, R, SQL
ML Frameworks: PyTorch, TensorFlow, Scikit-learn, JAX (working exposure)
LLM and Agentic AI: LangChain, vLLM, RAG, Multi-Agent Architectures, Agentic AI
Computer Vision: OpenCV, Homography, Affine Calibration, Perspective Geometry, Optimization-Based Path Planning
Research Areas: Multimodal Sensing, Applied Research, Evaluation and Benchmarking, Statistics, Linear Algebra, Optimization
Infra and Tools: Git, GitHub Actions, CI/CD, Linux CLI, SLURM, HPC, Pytest
Certifications: ML Specialization (Coursera/Stanford); Big Data and Spark (EdX)

PROFESSIONAL EXPERIENCE

AI Intern | Semine AS, Kristiansand, Norway — **Jan 2024 – Jun 2024**

- Designed and deployed a cloud-native document intelligence pipeline on Azure AI Studio integrating OCR and REST API field extraction, reducing manual processing time by 30% in production.
- Fine-tuned document models across 200+ foreign-language invoice types using transfer learning, boosting extraction accuracy by 15% over baseline through structured validation.
- Explored accelerating the document intelligence pipeline using JAX, gaining hands-on exposure to JIT compilation for numerical workloads.

SELECTED PROJECTS

CodeLens: RAG-Based Codebase Q&A System | Independent Project — **May 2026 – Present**
Self-directed project in ML systems engineering, built for hands-on depth in core AI infrastructure

- Shipped a RAG-based codebase intelligence API using hybrid BM25 and semantic retrieval with cross-encoder reranking, fixing a retrieval regression that raised hit rate from 85% to 100% on a hand-labeled 26-query evaluation set.
- Built a GitHub Actions CI pipeline gating merges on automated retrieval and LLM-judged quality checks, root-causing 3 distinct caching bugs that cut a 3.5-hour reindex step down to minutes on every run after the first.
- Achieved approximately 80% faithfulness and answer-relevancy scores on an automated LLM-judged quality gate, cutting judge token usage by approximately 33% (approximately 87K tokens per run) and generation cost by approximately 96%, bringing full evaluation cost down to $0.62 per run for deployment gating. *https://github.com/mvsg2/CodeLens*

Tau-Bench: Large-Scale LLM Agent Benchmarking | Arizona State University — **Jan 2026 – May 2026**

- Contributed 11 source-file changes to the open-source vLLM framework to enable distributed model serving on Intel Gaudi2 HPUs and NVIDIA A100 GPUs, benchmarking Qwen3 models (4B to 32B) across 825 total runs.
- Applied algorithmic strategies (ReAct, ACT, function-calling) across agent tasks and co-authored 115 SLURM scripts for fault-tolerant distributed execution at scale.
- Led error taxonomy over 3,677 failures; co-designed an LLM-as-judge architecture improving reliability by ~2-3%. *https://github.com/mvsg2/tau-bench-CSE598-repo*

Autonomous Maze-Solving Robot | Arizona State University — **Aug 2025 – Dec 2025**

- Built a computer vision pipeline using OpenCV (Otsu thresholding, contour detection, and perspective transform homography) to rectify an overhead camera image of any 4x4 maze into a top-down occupancy grid, applying projective geometry and linear algebra for coordinate mapping.
- Integrated an LLM via agentic tool calls for automated entrance/exit detection, then applied optimization-based path planning (BFS and A* with Manhattan distance heuristic) to compute minimum-length waypoint sequences for autonomous execution.
- Calibrated pixel-to-robot mapping via 6-parameter affine fit, solving mazes at any camera angle. *https://github.com/mvsg2/RAS-545-Midterm-Project*

Harmful Brain Activity Classification, EEG Pipeline | University of Agder — **Jan 2024 – Jun 2024**

- Trained and evaluated ResNet, EfficientNet, and MobileNet CNNs in PyTorch on NVIDIA V100 GPUs with transfer learning and cosine/step/exponential LR scheduling for multimodal EEG spectrogram sensing, using Scikit-learn for preprocessing and evaluation metrics.
- Achieved 71% validation accuracy across multiple harmful activity classes through statistical evaluation of model performance, with EfficientNet outperforming ResNet and MobileNet by approximately 5%, demonstrating transfer learning as an effective approach for clinical classification with limited labeled data.
- Developed a working prototype for an EEG-based diagnosis system using the higher-performing EfficientNet model.